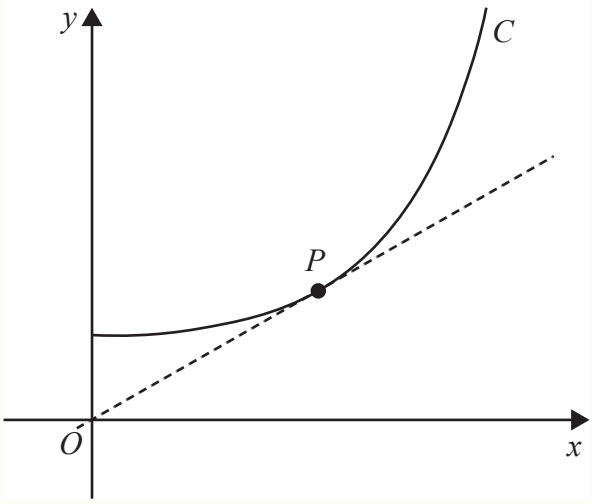



**Figure 3**

Figure 3 shows a sketch of part of the curve $C$ with equation

$$y = \sqrt{3 + 4e^{x^2}} \qquad x \geqslant 0$$

(a) Find $\dfrac{dy}{dx}$, giving your answer in simplest form.

**(2)**

The point $P$ with $x$ coordinate $\alpha$ lies on $C$.

Given that the tangent to $C$ at $P$ passes through the origin, as shown in Figure 3,

(b) show that $x = \alpha$ is a solution of the equation

$$4x^2e^{x^2} - 4e^{x^2} - 3 = 0$$

**(3)**

(c) Hence show that $\alpha$ lies between 1 and 2

**(2)**

(d) Show that the equation in part (b) can be written in the form

$$x = \frac{1}{2}\sqrt{4 + 3e^{-x^2}}$$

**(1)**

The iteration formula

$$x_{n+1} = \frac{1}{2}\sqrt{4 + 3e^{-x_n^2}}$$

with $x_1 = 1$ is used to find an approximation for $\alpha$.

(e) Use the iteration formula to find, to 4 decimal places, the value of

(i) $x_3$

(ii) $\alpha$